Insurance Claim Prediction Model

Objective:

The objective of this project is to develop a statistical model to predict the probability of insurance claims using policyholder demographics, driving behavior, and vehicle characteristics. This project applies actuarial modeling techniques to assess risk.

Dataset:

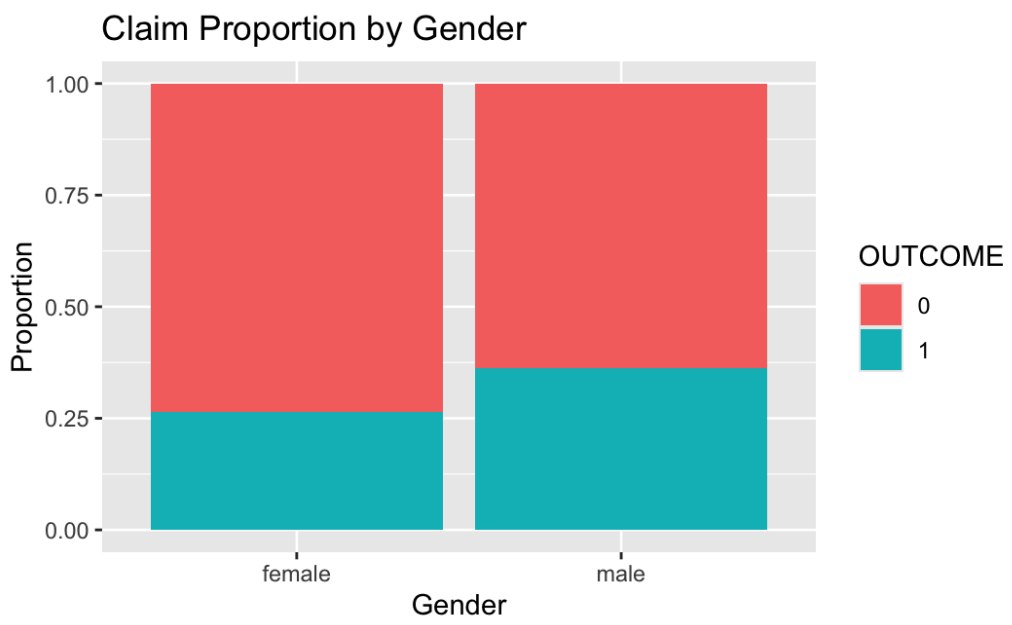
This project used a dataset containing 10000 observations with the following variables:

- Demographics: Age, Gender, Marital Status, Income
- Driving behavior: driving experience, speeding violations, DUIs
- Vehicle characteristics: vehicle type, vehicle year
- Other: annual mileage, past accidents
- The target variable indicates whether a claim occurred (1 = claim, 0 = no claim)

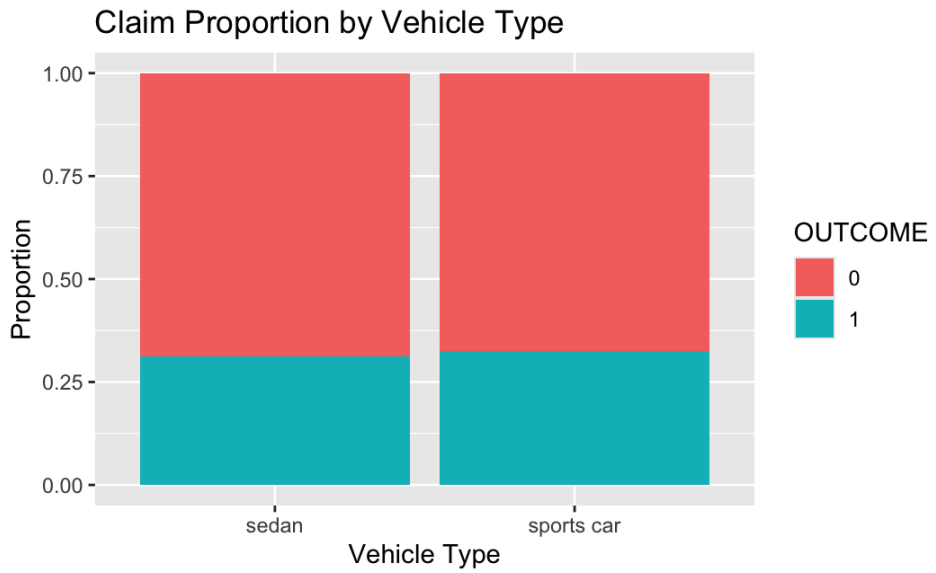
Missing data points for credit score and annual mileage were filled using column medians.

Exploratory analysis:

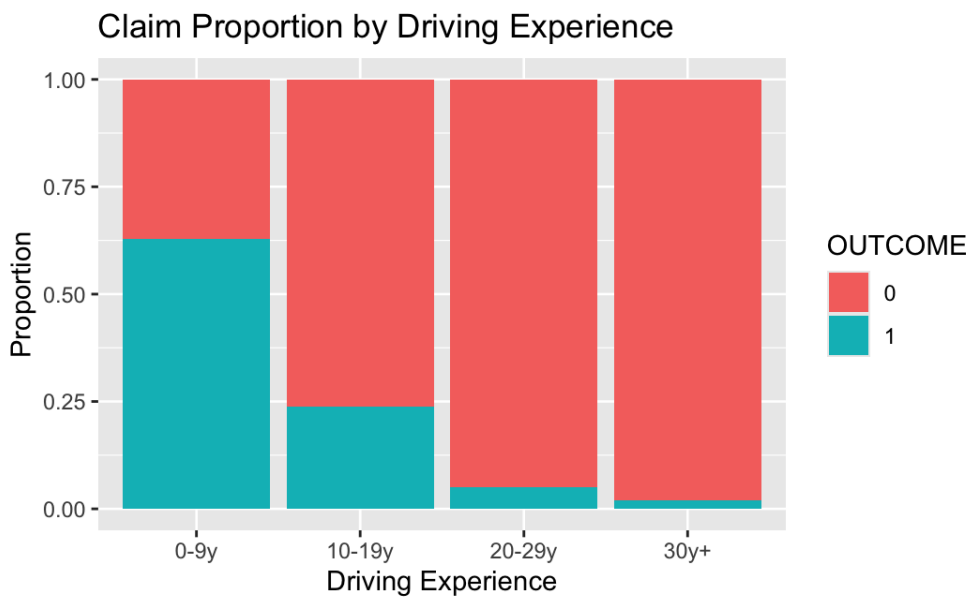
Initial analysis compared claim rates across key variables: driving experience, gender, and vehicle type.



Male drivers showed higher claim proportions



There was no significant difference observed between the claim proportions from sedans and sports cars.



There was a significant difference observed between claim proportions from different levels of driving experience, with less experienced drivers having significantly higher claim rates.

Model:

A logistic regression model was used to predict claim probability:

- Response: claim occurrence (0 = no claim, 1 = claim)
- Method: generalized linear model (logit link)
- Predictors: demographic, behavioral and vehicle variables

Significant predictors ($p < 0.05$)

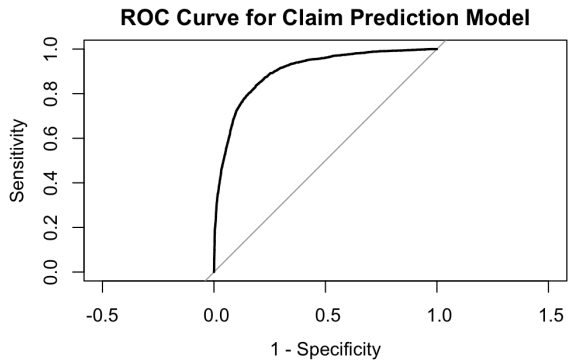
Predictor	Odds Ratio	Interpretation
Experience: 10-19y vs 0-9y	0.18	82% lower claim odds
Experience: 20-29y vs 0-9y	0.04	96% lower claim odds
Experience: 30y+ vs 0-9y	0.02	98% lower claim odds
Gender: male vs female	2.56	2.56x higher claim odds
Vehicle owned	0.19	81% lower claim odds
Vehicle year: post-2015	0.19	82% lower claim odds
Married vs single	0.65	35% lower claim odds
Children	0.84	16% lower per child
Annual mileage	1.000043	+54% per 10,000 miles
Speeding violations	1.09	+8.6% per violation
Past accidents	0.8	20% lower claim odds

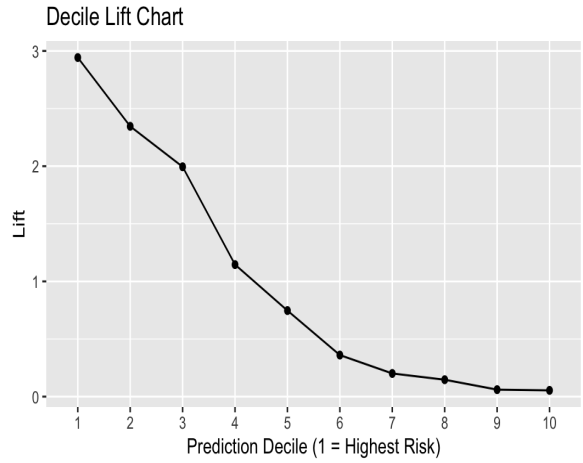
Age, income, education, race, credit score, vehicle type, and DUIs were not independently significant when controlling for other variables (all $p > 0.05$). Notably, past accidents show a negative coefficient, contrary to intuition. This is likely due to survivorship bias: high-accident drivers may have been cancelled or priced out, leaving only lower-risk high accident policyholders in the dataset.

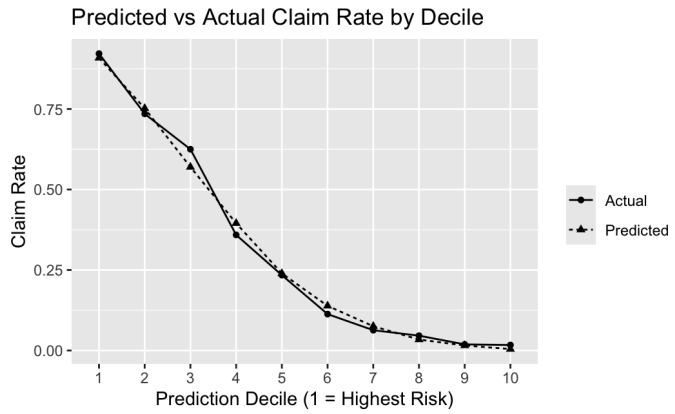
Model Evaluation:

The accuracy of the model was assessed using:

- Area Under Curve (AUC): measured classification performance.
- AIC: evaluated model fit.
- Decile Lift Analysis: confirmed strong ranking of high-risk policyholders
- Calibration Plot: compared predicted vs actual claim probabilities







Model Fit Metrics

Metric	Value	Interpretation
AUC	0.900	Model correctly ranks 90% of claim/no-claim pairs
Accuracy (threshold = 0.5)	84.3%	At a 0.5 probability threshold, reflects strong overall classification
AIC	7375	Benchmark for comparing alternative model specifications
McFadden R^2	0.41	Strong fit

Null deviance	12434	Deviance with intercept only
Residual deviance	7327	41% reduction from null

Conclusion:

This logistic regression model identifies the key drivers of claim risk in a personal auto portfolio. Driving experience proved to be the dominant predictor, with drivers with 30+ years of experience yielding 98% lower claim odds than the least experienced group with 0-9 years of experience. Vehicle ownership and vehicle year (post 2015) are also strong predictors, each reducing claim odds by 81-82%.

The findings from this project can be applied to actuarial rating relativities. For instance, a male driver with under 9 years of experience and 2 speeding violations would yield a combined frequency relativity of approximately 3.05 relative to the base policyholder.